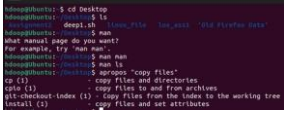
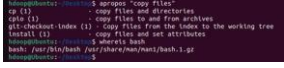
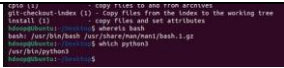
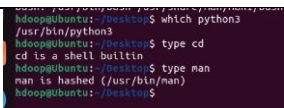


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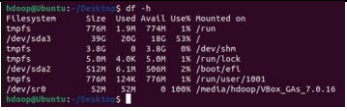
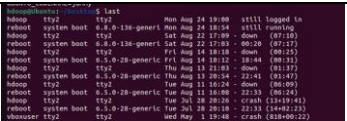
1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man [command]	man ls	
To get manual page for the unknown command	apropos or man -k	apropos [keyword]	apropos "copy files"	
To know the source file binary	whereis	whereis [command]	whereis bash	
To know the path of the command	which	which [command]	which python3	
To know the command is external or internal	type	type [command]	type cd	

To get help for the internal command	help	help [command]	help pwd	<pre> hadoop@ubuntu:~/hadoop\$ help pwd pwd [options] Print the name of the current working directory. Options: -L print the value of \$PWD if it names the current working directory -P print the physical directory, without any symbolic links By default, 'pwd' behaves as if '-L' were specified. Exit Status: Returns 0 unless an invalid option is given or the current directory cannot be read. </pre>
To list out bash commands	compgen	compgen [option]	compgen -b	<pre> hadoop@Ubuntu: ~/Desktop\$ compgen -b _ [alias bg bind break builtin caller cd command compgen complete </pre>
To know the usage of the command	--help	[command] --help	mkdir --help	<pre> hadoop@ubuntu:~/hadoop\$ mkdir --help Usage: mkdir [OPTION]... DIRECTORY... Create the DIRECTORY(ies), if they do not already exist. Mandatory arguments to long options are mandatory for short options too. -m, --mode=MODE set file mode (as in chmod), not -perm, --perm -p, --parents no error if existing, make parent directories as needed -v, --verbose print a message for each created directory -Z, --security set SELinux security context of each created directory to the default type -t, --target=CTX like -p, or if CTX is specified then set the SELinux or SMAC security context to CTX -help display this help and exit -version output version information and exit See documentation online help: <https://www.gnu.org/software/coreutils/> Full documentation at: <https://www.gnu.org/software/coreutils/mkdir/> or available locally via: 'info (coreutils) mkdir invocation' </pre>

2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	<pre> hadoop@Ubuntu: ~/Desktop\$ date Monday 24 August 2026 07:49:22 PM IST hadoop@Ubuntu: ~/Desktop\$ </pre>
To print calendar	cal	cal [month] [year]	cal	<pre> hadoop@Ubuntu: ~/Desktop\$ cal 2026 August Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 hadoop@Ubuntu: ~/Desktop\$ </pre>
To print kernel version	uname	uname -r	uname -r	<pre> hadoop@Ubuntu: ~/Desktop\$ uname Linux hadoop@Ubuntu: ~/Desktop\$ uname -r 6.8.0-136-generic hadoop@Ubuntu: ~/Desktop\$ </pre>

To know free space in the system	df	df [options]	df -h	
To know about the Linux release	cat /lsb_release	cat /etc/os-release	cat /etc/os-release	
To show login history	last	last	last	
To check the current user privileges	sudo or id	sudo -l or id	sudo -l or id	
To find about the Linux release	cat /lsb_release	cat /etc/os-release	cat /etc/os-release	
To check system uptime	uptime	uptime	uptime	
To shutdown	shutdown	shutdown [time]	shutdown now	
To restart	reboot	reboot	reboot	

3. Navigation

Task	Command	Syntax	Screenshots
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Task	Syntax	Command																																																								
How to identify the file system	df	<pre>hdoop@Ubuntu:~\$ cd Desktop hdoop@Ubuntu:~/Desktop\$ cd / hdoop@Ubuntu:/\$ df -T</pre> <table><thead><tr><th>Filesystem</th><th>Type</th><th>1K-blocks</th><th>Used</th><th>Available</th><th>Use%</th><th>Mounted on</th></tr></thead><tbody><tr><td>tmpfs</td><td>tmpfs</td><td>793808</td><td>1616</td><td>792192</td><td>1%</td><td>/run</td></tr><tr><td>/dev/sda3</td><td>ext4</td><td>40453376</td><td>20133600</td><td>18232660</td><td>53%</td><td>/</td></tr><tr><td>tmpfs</td><td>tmpfs</td><td>3969032</td><td>0</td><td>3969032</td><td>0%</td><td>/dev/shm</td></tr><tr><td>tmpfs</td><td>tmpfs</td><td>5120</td><td>4</td><td>5116</td><td>1%</td><td>/run/lock</td></tr><tr><td>/dev/sda2</td><td>vfat</td><td>524252</td><td>6232</td><td>518020</td><td>2%</td><td>/boot/efi</td></tr><tr><td>tmpfs</td><td>tmpfs</td><td>793804</td><td>116</td><td>793688</td><td>1%</td><td>/run/user/1001</td></tr><tr><td>/dev/sr0</td><td>iso9660</td><td>52274</td><td>52274</td><td>0</td><td>100%</td><td>/media/hdoop/VBox_GAs_7.0.16</td></tr></tbody></table> <pre>hdoop@Ubuntu:/\$</pre> df -T	Filesystem	Type	1K-blocks	Used	Available	Use%	Mounted on	tmpfs	tmpfs	793808	1616	792192	1%	/run	/dev/sda3	ext4	40453376	20133600	18232660	53%	/	tmpfs	tmpfs	3969032	0	3969032	0%	/dev/shm	tmpfs	tmpfs	5120	4	5116	1%	/run/lock	/dev/sda2	vfat	524252	6232	518020	2%	/boot/efi	tmpfs	tmpfs	793804	116	793688	1%	/run/user/1001	/dev/sr0	iso9660	52274	52274	0	100%	/media/hdoop/VBox_GAs_7.0.16
Filesystem	Type	1K-blocks	Used	Available	Use%	Mounted on																																																				
tmpfs	tmpfs	793808	1616	792192	1%	/run																																																				
/dev/sda3	ext4	40453376	20133600	18232660	53%	/																																																				
tmpfs	tmpfs	3969032	0	3969032	0%	/dev/shm																																																				
tmpfs	tmpfs	5120	4	5116	1%	/run/lock																																																				
/dev/sda2	vfat	524252	6232	518020	2%	/boot/efi																																																				
tmpfs	tmpfs	793804	116	793688	1%	/run/user/1001																																																				
/dev/sr0	iso9660	52274	52274	0	100%	/media/hdoop/VBox_GAs_7.0.16																																																				

5. Globbing

Create a dedicated practice directory containing sample files such as: index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt Use **globbing patterns wherever possible**, rather than specifying filenames individually. a. List all files ending in .html.

- List all files **other than .html files** using extended globbing.
- List filenames containing at least one numeric digit.
- List files whose names begin with user followed by a single digit.
- List files ending in either .txt or .log using a single **extended glob pattern**.
- List files that do **not** end in .txt or .log.
- List files beginning with either access or error using extended globbing.
- List .html files whose filenames contain a numeric digit.

```

hdoop@Ubuntu:~$ cd Desktop
hdoop@Ubuntu:~/Desktop$ mkdir practice
hdoop@Ubuntu:~/Desktop$ cd practice
hdoop@Ubuntu:~/Desktop/practice$ touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt
hdoop@Ubuntu:~/Desktop/practice$ ls
access.log admin.htm backup1.html backup2.txt error.log file123 index.html login.html report.txt user1.txt user2.log
hdoop@Ubuntu:~/Desktop/practice$ ls *.html
backup1.html index.html login.html
hdoop@Ubuntu:~/Desktop/practice$ ls !(*.html)
access.log admin.htm backup2.txt error.log file123 report.txt user1.txt user2.log
hdoop@Ubuntu:~/Desktop/practice$ ls *[0-9]*
backup1.html backup2.txt file123 user1.txt user2.log
hdoop@Ubuntu:~/Desktop/practice$ ls user[0-9]*
user1.txt user2.log
hdoop@Ubuntu:~/Desktop/practice$ ls *.*(txt|log)
access.log backup2.txt error.log report.txt user1.txt user2.log
hdoop@Ubuntu:~/Desktop/practice$ ls !(*.txt|*.log)
admin.htm backup1.html file123 index.html login.html
hdoop@Ubuntu:~/Desktop/practice$ ls @(access|error)*
access.log error.log
hdoop@Ubuntu:~/Desktop/practice$ ls *[0-9]*.html
backup1.html
hdoop@Ubuntu:~/Desktop/practice$

```

6. Extended Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

```

hdoop@Ubuntu:~/Desktop$ cd Assignment2
hdoop@Ubuntu:~/Desktop/Assignment2$ ls
sample_dir  sample.txt
hdoop@Ubuntu:~/Desktop/Assignment2$ touch access.log error.log auth.log auth.log.1 auth.log.2 system.log notes.txt
hdoop@Ubuntu:~/Desktop/Assignment2$ ls
access.log  auth.log  auth.log.1  auth.log.2  error.log  notes.txt  sample_dir  sample.txt  system.log
hdoop@Ubuntu:~/Desktop/Assignment2$ ls @(access|error|auth).log
access.log  auth.log  error.log
hdoop@Ubuntu:~/Desktop/Assignment2$ ls !(notes.txt)
access.log  auth.log  auth.log.1  auth.log.2  error.log  sample.txt  system.log

sample_dir:
new_file.txt
hdoop@Ubuntu:~/Desktop/Assignment2$ ls auth.log.@(1|2)
auth.log.1  auth.log.2
hdoop@Ubuntu:~/Desktop/Assignment2$ ls auth.log.[12]
auth.log.1  auth.log.2
hdoop@Ubuntu:~/Desktop/Assignment2$

```

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

```

hdoop@Ubuntu:~/Desktop/Assignment2$ mkdir extglob
hdoop@Ubuntu:~/Desktop/Assignment2$ cd extglob
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ touch malware1.log malware2.log malwareA.log malware_report.txt normal.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls
malware1.log  malware2.log  malwareA.log  malware_report.txt  normal.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls malware*[0-9]*.log
malware1.log  malware2.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$

```

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log use extended globbing to select

.log files containing either failed_login or successful_login.

Construct a pattern that lists all .log files **except** firewall.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ touch ailed_login.log successful_login.log login_backup.log login.txt firewall.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls
ailed_login.log  firewall.log  login_backup.log  login.txt  successful_login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ mv ailed_login.log failed_login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls
failed_login.log  firewall.log  login_backup.log  login.txt  successful_login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls @(failed_login|successful_login).log
failed_login.log  successful_login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ ls !(firewall).log
failed_login.log  login_backup.log  successful_login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

7. Filters

a. Display the first **10 entries** of login.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 10 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
Aug 15 09:01:07 server sshd[2102]: Failed password for student2 from 198.51.100.44
Aug 15 09:02:14 server sshd[2103]: Failed password for analyst from 203.0.113.77
Aug 15 09:03:21 server sshd[2104]: Accepted password for admin from 192.168.10.45
Aug 15 09:04:28 server sshd[2105]: Accepted password for root from 192.168.10.21
Aug 15 09:05:35 server sshd[2106]: Failed password for guest from 198.51.100.44
Aug 15 09:06:42 server sshd[2107]: Failed password for student1 from 203.0.113.77
Aug 15 09:07:49 server sshd[2108]: Accepted password for student2 from 192.168.10.45
Aug 15 09:08:56 server sshd[2109]: Accepted password for analyst from 192.168.10.21
Aug 15 09:09:03 server sshd[2110]: Failed password for admin from 198.51.100.44
```

b. Display the first **5 entries** of access.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 5 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
```

c. Display the first **15 security events** from security.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 15 security.log
2026-08-15 09:00:00 INFO user login successful; source=192.168.10.21; event_id=SEC0001
2026-08-15 09:00:29 WARN failed login attempt; source=192.168.10.22; event_id=SEC0002
2026-08-15 09:00:58 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0003
2026-08-15 09:01:27 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0004
2026-08-15 09:01:56 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0005
2026-08-15 09:02:25 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0006
2026-08-15 09:02:54 INFO user logout; source=203.0.113.77; event_id=SEC0007
2026-08-15 09:03:23 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0008
2026-08-15 09:03:52 INFO user login successful; source=192.168.10.21; event_id=SEC0009
2026-08-15 09:04:21 WARN failed login attempt; source=192.168.10.22; event_id=SEC0010
2026-08-15 09:04:50 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0011
2026-08-15 09:05:19 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0012
2026-08-15 09:05:48 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0013
2026-08-15 09:06:17 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0014
2026-08-15 09:06:46 INFO user logout; source=203.0.113.77; event_id=SEC0015
```

d. Display the first **20 events** recorded in incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 20 incident.log
2026-08-15 09:00:00 INFO login success; source=192.168.10.21; incident_event=001
2026-08-15 09:00:13 INFO web request; source=192.168.10.22; incident_event=002
2026-08-15 09:00:26 INFO logout; source=192.168.10.30; incident_event=003
2026-08-15 09:00:39 INFO scheduled scan; source=192.168.10.45; incident_event=004
```

e. Display only the **first entry** in login.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 1 login.log
Aug 15 09:00:00 server sshd[2101]: Accepted password for student1 from 192.168.10.21
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

f. Display the first **25 web requests** recorded in access.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 25 access.log
192.168.10.21 - - [15/Aug/2026:09:00:00 +0530] "GET / HTTP/1.1" 200 512
192.168.10.22 - - [15/Aug/2026:09:00:37 +0530] "POST /login HTTP/1.1" 200 649
192.168.10.30 - - [15/Aug/2026:09:01:14 +0530] "GET /dashboard HTTP/1.1" 302 786
192.168.10.45 - - [15/Aug/2026:09:01:51 +0530] "GET /admin HTTP/1.1" 403 923
203.0.113.25 - - [15/Aug/2026:09:02:28 +0530] "GET /robots.txt HTTP/1.1" 404 1060
198.51.100.44 - - [15/Aug/2026:09:03:05 +0530] "POST /api/status HTTP/1.1" 401 1197
```

- g. Display the last **10 entries** of login.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 10 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
Aug 15 09:44:08 server sshd[2145]: Accepted password for analyst from 192.168.10.21
Aug 15 09:45:15 server sshd[2146]: Failed password for admin from 198.51.100.44
```

- h. Display the last **5**

entries of access.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 5 access.log
192.168.10.45 - - [15/Aug/2026:09:46:15 +0530] "GET /api/status HTTP/1.1" 403 2595
203.0.113.25 - - [15/Aug/2026:09:46:52 +0530] "GET /assets/app.js HTTP/1.1" 404 2732
198.51.100.44 - - [15/Aug/2026:09:47:29 +0530] "POST / HTTP/1.1" 401 2869
203.0.113.77 - - [15/Aug/2026:09:48:06 +0530] "GET /login HTTP/1.1" 200 3006
198.51.100.91 - - [15/Aug/2026:09:48:43 +0530] "GET /dashboard HTTP/1.1" 200 3143
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

- i. Display the **20 most recent security events** from security.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 20 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
2026-08-15 09:50:16 INFO user login successful; source=192.168.10.21; event_id=SEC0105
```

- j. Display the **25 most recent events** from incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 25 incident.log
2026-08-15 10:42:55 INFO login success; source=192.168.10.45; incident_event=476
2026-08-15 10:43:08 INFO web request; source=203.0.113.25; incident_event=477
2026-08-15 10:43:21 INFO logout; source=198.51.100.44; incident_event=478
2026-08-15 10:43:34 INFO scheduled scan; source=203.0.113.77; incident_event=479
```

- k. Display only the **last entry** in security.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 1 security.log
2026-08-15 09:57:31 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0120
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

- l. Display the last **50 events** recorded in incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 25 incident.log
2026-08-15 10:42:55 INFO login success; source=192.168.10.45; incident_event=476
2026-08-15 10:43:08 INFO web request; source=203.0.113.25; incident_event=477
```

- m. Display the contents of login.log starting from **line 41 until the end of the file**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n +41 login.log
Aug 15 09:40:40 server sshd[2141]: Accepted password for root from 192.168.10.21
Aug 15 09:41:47 server sshd[2142]: Failed password for guest from 198.51.100.44
Aug 15 09:42:54 server sshd[2143]: Failed password for student1 from 203.0.113.77
Aug 15 09:43:01 server sshd[2144]: Accepted password for student2 from 192.168.10.45
```

- n. Display the contents of security.log starting from **line 101 until the end of the file**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n +101 security.log
2026-08-15 09:48:20 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0101
2026-08-15 09:48:49 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0102
2026-08-15 09:49:18 INFO user logout; source=203.0.113.77; event_id=SEC0103
2026-08-15 09:49:47 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0104
```

- o. Find the total number of lines in login.log.
- p. Find the total number of lines in access.log.
- q. Find the total number of security events in security.log.
- r. Determine the total number of events recorded in incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -l login.log
50 login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -l access.log
80 access.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -l security.log
120 security.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -l incident.log
500 incident.log
```

- s. Count the number of words in login.log.

- t. Count the number of bytes in security.log.
- u. Display the **line, word, and byte counts** of access.log.
- v. Find the line counts of all four .log files using a single wc command.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -w login.log
550 login.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -c security.log
10875 security.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc access.log
 80  800 6414 access.log
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ wc -l login.log access.log security.log incident.log
 50 login.log
 80 access.log
120 security.log
500 incident.log
750 total
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

- w. Display the first **20 entries** of login.log and, from those entries, display only the **last 5**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 20 login.log | tail -n 5
Aug 15 09:15:45 server sshd[2116]: Accepted password for admin from 192.168.10.45
Aug 15 09:16:52 server sshd[2117]: Accepted password for root from 192.168.10.21
```

- x. Display the last **20 entries** of login.log and, from those entries, display only the **first 5**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 20 login.log | head -n 5
Aug 15 09:30:30 server sshd[2131]: Failed password for student1 from 203.0.113.77
Aug 15 09:31:37 server sshd[2132]: Accepted password for student2 from 192.168.10.45
```

- y. Display the first **30 entries** of access.log and then display only the **last 10 entries** from that output.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 30 access.log | tail -n 10
203.0.113.25 - - [15/Aug/2026:09:12:20 +0530] "GET /assets/app.js HTTP/1.1" 302 3252
198.51.100.44 - - [15/Aug/2026:09:12:57 +0530] "POST / HTTP/1.1" 403 3389
```

z. Display the last **30 entries** of access.log and then display only the **first 10 entries**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 30 access.log | head -n 10
192.168.10.30 - - [15/Aug/2026:09:30:50 +0530] "GET /login HTTP/1.1" 302 3266
192.168.10.45 - - [15/Aug/2026:09:31:27 +0530] "GET /dashboard HTTP/1.1" 403 3403
```

aa.Display the first **50 events** from security.log and then display the **last 10 events** from those 50.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 50 security.log | tail -n 10
2026-08-15 09:19:20 INFO user login successful; source=192.168.10.21; event_id=SEC0041
2026-08-15 09:19:49 WARN failed login attempt; source=192.168.10.22; event_id=SEC0042
```

bb.Display the last **50 events** from security.log and then display the **first 10 events** from those 50.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 50 security.log | head -n 10
2026-08-15 09:33:50 INFO user logout; source=203.0.113.77; event_id=SEC0071
2026-08-15 09:34:19 WARN access denied to restricted resource; source=198.51.100.91; event_id=SEC0072
```

cc. Display entries from **line 11 to line 20** of login.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 20 login.log | tail -n 10
Aug 15 09:10:10 server sshd[2111]: Failed password for root from 203.0.113.77
Aug 15 09:11:17 server sshd[2112]: Accepted password for guest from 192.168.10.45
```

dd.Display entries from **line 21 to line 30** of login.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 30 login.log | tail -n 10
Aug 15 09:20:20 server sshd[2121]: Accepted password for analyst from 192.168.10.21
Aug 15 09:21:27 server sshd[2122]: Failed password for admin from 198.51.100.44
```

ee. Display entries from **line 11 to line 20** of access.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 20 access.log | tail -n 10
192.168.10.30 - - [15/Aug/2026:09:06:10 +0530] "GET /admin HTTP/1.1" 404 1882
192.168.10.45 - - [15/Aug/2026:09:06:47 +0530] "GET /robots.txt HTTP/1.1" 401 2019
```

ff. Display entries from **line 31 to line 40** of access.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 40 access.log | tail -n 10
203.0.113.77 - - [15/Aug/2026:09:18:30 +0530] "GET /dashboard HTTP/1.1" 200 526
198.51.100.91 - - [15/Aug/2026:09:19:07 +0530] "GET /admin HTTP/1.1" 200 663
```

gg. Display security events from **line 21 to line 30** of security.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 30 security.log | tail -n 10
2026-08-15 09:09:40 WARN repeated authentication failure; source=203.0.113.25; event_id=SEC0021
2026-08-15 09:10:09 ALERT possible port scan detected; source=198.51.100.44; event_id=SEC0022
```

hh. Display security events from **line 51 to line 60** of security.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 60 security.log | tail -n 10
2026-08-15 09:24:10 ALERT firewall blocked connection; source=192.168.10.30; event_id=SEC0051
2026-08-15 09:24:39 INFO antivirus scan completed; source=192.168.10.45; event_id=SEC0052
```

ii. Display incident events from **line 101 to line 110** of incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 110 incident.log | tail -n 10
2026-08-15 09:21:40 INFO login success; source=203.0.113.25; incident_event=101
2026-08-15 09:21:53 INFO web request; source=198.51.100.44; incident_event=102
```

jj. Display incident events from **line 201 to line 220** of incident.log.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 220 incident.log | tail -n 20
2026-08-15 09:43:20 WARNING failed SSH login; source=192.168.10.21; incident_event=201
2026-08-15 09:43:33 ALERT firewall block; source=192.168.10.22; incident_event=202
```

kk. Display incident events from **line 491 to line 500**.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 500 incident.log | tail -n 10
2026-08-15 10:46:10 INFO login success; source=192.168.10.30; incident_event=491
2026-08-15 10:46:23 INFO web request; source=192.168.10.45; incident_event=492
```

ll. Take the first **20 entries** of login.log and count how many lines are displayed.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 20 login.log | wc -l
20
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

mm. Take the last **15 entries** of access.log and count how many lines are displayed.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 15 access.log | wc -l
15
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

nn. Take the first **50 entries** of security.log and count the number of words in those entries.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 50 security.log | wc -w
412
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

oo. Take the last **25 entries** of security.log and count the number of lines.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ tail -n 25 security.log | wc -l
25
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

pp. Extract lines **21–30** from login.log and count the number of extracted lines.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 30 login.log | tail -n 10 | wc -l
10
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

qq. Extract lines **51–60** from access.log and count the number of extracted lines.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 60 access.log | tail -n 10 | wc -l
10
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

rr. Extract events **101–150** from incident.log and count how many events are displayed.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 150 incident.log | tail -n 50 | wc -l
50
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```

ss. Display the first **100 events** from incident.log, take the last **20 events** from that output, and count the number of lines.

```
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$ head -n 100 incident.log | tail -n 20 | wc -l
20
hdoop@Ubuntu:~/Desktop/Assignment2/extglob$
```